Fundamentals of Machine Learning
Hard Mock Midterm (Lectures 1 to 5)

Two hours, 100 points, no documents. New situations, and results the lectures never derived: you prove them from the tools you have

Fundamentals of Machine Learning

2026-10-30

Central question. The lectures derived a handful of results: the Bayes rule, the cost threshold, cross-entropy as maximum likelihood, the bias–variance decomposition, the LDA score, the Newton leaf. Can you use the same few moves (condition on x, take an expectation, differentiate, apply Bayes' rule, count) to derive results you have never seen?

Instructions. Every problem contains at least one statement you are asked to *show*. None of these results is in the lecture notes; each follows in a few lines from something that is. Hints give the one identity you might not remember. State assumptions. A correct derivation with an arithmetic slip earns most of the marks; a correct number without the derivation earns few.

1. Fraud with amount-dependent costs and labels that lie — 20 minutes — [16 pts]

A card issuer scores each transaction x with $\eta(x) = \mathbb{P}(Y = 1 \mid x)$, where $Y = 1$ is fraud. Blocking a legitimate transaction costs $c = 5$ (support call, lost goodwill). Letting a fraud through costs its amount $A(x)$, which is known at decision time.

(a) Derive the optimal rule and show that it is a threshold $t(x)$ that depends on x. Compute it for a 20-euro and a 500-euro transaction. Explain why the set of blocked transactions is no longer a level set of η, and what that implies for anyone who reports "the" threshold of this system. [4 pts]

(b) Fraud labels come from chargebacks. A fraud is disputed (labelled 1) with probability $1-\rho_1$ and silently absorbed (labelled 0) with probability ρ_1; a legitimate transaction is wrongly disputed with probability ρ_0. The flips depend on the true class only, not on x. Let $\tilde{Y}$ be the recorded label and $\tilde{\eta}(x) = \mathbb{P}(\tilde{Y} = 1 \mid x)$. Show that

$$\tilde{\eta}(x) = \rho_0 + (1 - \rho_0 - \rho_1)\,\eta(x).$$

[4 pts]

(c) A model trained with cross-entropy on the recorded labels is well calibrated for $\tilde{Y}$. You want the decision of part (a) with $t = 0.2$. Derive the threshold $\tilde{t}$ to apply to the model's output, and compute it for $\rho_0 = 0.01$, $\rho_1 = 0.30$. [3 pts]

(d) Show that if $\rho_0 + \rho_1 < 1$, ranking transactions by $\tilde{\eta}$ gives the same order as ranking by η. Which evaluation quantities are therefore unaffected by this label noise, and which are not? What happens when $\rho_0 + \rho_1 = 1$? [3 pts]

(e) Show that with symmetric noise $\rho_0 = \rho_1 = \rho < \frac{1}{2}$ the zero–one Bayes classifier learned from noisy labels is the same as the clean one, and explain why the same is false for the cost-sensitive rule. [2 pts]

2. Which losses learn probabilities? 20 minutes [16 pts]

Binary $Y \in \{0,1\}$, $\eta = \mathbb{P}(Y=1 \mid x)$, and a model outputs a number $q \in [0,1]$ at x. A loss $\ell(y,q)$ is called *proper* if $q=\eta$ minimises the conditional risk $\mathbb{E}[\ell(Y,q) \mid x]$ for every η.

(a) Brier loss $\ell(y,q)=(y-q)^2$. Show that $\mathbb{E}[(Y-q)^2 \mid x] = \eta(1-\eta)+(\eta-q)^2$. Conclude that it is proper, and name the two terms. [3 pts]

(b) Absolute loss $\ell(y,q)=|y-q|$. Compute $\mathbb{E}[|Y-q| \mid x]$, find its minimiser over $[0,1]$, and conclude that it is not proper. What does a model trained with it output, and for which decision problems is that harmless and for which is it fatal? Use $\eta = 0.3$ and the cost threshold 0.2 as your example. [4 pts]

(c) Explain in two lines, with a result from the course, why cross-entropy is proper. [1 pts]

(d) To "fix imbalance", a team trains with class weights: every positive example's cross-entropy is multiplied by w_1, every negative's by w_0. Show that the conditional risk is minimised by

$$q^* = \frac{w_1\,\eta}{w_1\,\eta + w_0\,(1-\eta)}, \qquad \text{equivalently} \qquad \frac{q^*}{1-q^*} = \frac{w_1}{w_0}\cdot\frac{\eta}{1-\eta}.$$

[4 pts]

(e) The attacks are 5% of the training data and the team used $w_1=19$, $w_0=1$. The weighted model outputs $q=0.6$ for a flow. Recover η. Then explain which formula of the course this correction is identical to, and why. [4 pts]

3. Shrinking on purpose, and how large a validation set must be 25 minutes [18 pts]

Part I. A SOC wants the mean μ of daily alerts on a new sensor. It has n i.i.d. days $X_1,\dots,X_n$ with mean μ and variance σ^2, and considers the shrunken estimator $\hat{\mu}_c = c\,\bar{X}$ with $0 \le c \le 1$.

(a) Derive the bias, the variance and the mean squared error of $\hat{\mu}_c$. [3 pts]

(b) Show that the MSE is minimised at $c^* = \dfrac{\mu^2}{\mu^2+\sigma^2/n}$, that $c^*<1$ whenever $\sigma^2>0$, and that $c^*\to 1$ as $n\to\infty$. Interpret each of the three facts with the vocabulary of Lecture 3. [4 pts]

(c) For $\mu=2$, $\sigma^2=16$, $n=4$, compare the MSE of the unbiased estimator and of $\hat{\mu}_{c^*}$. Since c^* depends on the unknown μ, what is c in practice, and which object of Lecture 3 plays exactly this role for linear models? [3 pts]

Part II. For a loss in $[0,1]$ and n i.i.d. validation examples, Hoeffding's inequality gives

$$\mathbb{P}\big(|\widehat{R}_{\text{val}} - R| > \varepsilon\big) \le 2\,e^{-2n\varepsilon^2}.$$

(d) How many validation examples guarantee, with probability at least 0.95, that one fixed model's validation error is within 0.01 of its true error? [2 pts]

(e) The team now compares $m=100$ configurations on the same validation set and keeps the best. Using $\mathbb{P}(\bigcup_j A_j) \le \sum_j \mathbb{P}(A_j)$, show that *all* m validation errors are within ε of their true errors with probability at least $1-2m\,e^{-2n\varepsilon^2}$, derive the required n, and compute it. Explain why this is the quantitative version of a warning from the course. [4 pts]

(f) The validation set contains whole attack episodes of 200 near-identical flows each. Which hypothesis of the inequality fails, and does the true guarantee get better or worse than the formula says? [2 pts]

4. Counting failed logins, and data that are too easy — 25 minutes — [20 pts]

Part I. Let $X \in \{0, 1, 2, \ldots\}$ be the number of failed logins on a server in one minute. A generative detector assumes Poisson class-conditionals

$$\mathbb{P}(X = k \mid y = c) = \frac{\lambda_c^k \, e^{-\lambda_c}}{k!}, \qquad c \in \{0, 1\}.$$

(a) From a training sample with N_c minutes of class c, derive the maximum-likelihood estimate of λ_c. [3 pts]

(b) Derive the log posterior odds $\log \frac{\mathbb{P}(y=1|k)}{\mathbb{P}(y=0|k)}$ and show that it is linear in k. Give the decision boundary k^* under zero–one loss. [4 pts]

(c) With $\lambda_0 = 0.5$, $\lambda_1 = 6$, $\pi_1 = 0.02$: compute k^*, the alert rule, and the posterior for $k = 3$ and $k = 4$. [3 pts]

(d) Show that this posterior is exactly a logistic regression in k and give its weight and bias. Then: with 30 attack minutes and 30,000 normal ones, which route would you train, and which route degrades first if attacks arrive in bursts (variance of the count far above its mean)? [3 pts]

Part II. A logistic regression $\hat{p}_\theta(x) = \sigma(\theta^\top x)$ is trained by minimising the average cross-entropy, with no penalty. The training set is *linearly separable*: there is a w with $w^\top x^{(i)} > 0$ for every $y^{(i)} = 1$ and $w^\top x^{(i)} < 0$ for every $y^{(i)} = 0$.

(e) Show that the training loss of $\theta = a\,w$ is strictly decreasing in $a > 0$ and tends to 0 as $a \to \infty$. Conclude that the minimum is not attained and describe what gradient descent does. [4 pts]

(f) Show that adding $\lambda\|\theta\|_2^2$ with $\lambda > 0$ restores a minimiser, and state the consequence of the unpenalised fit for the probabilities a SOC analyst would see. [3 pts]

Hint. for (e), each term is $-\log \sigma(a\,w^\top x)$ or $-\log(1 - \sigma(a\,w^\top x))$; look at the sign of $w^\top x$. For (f), a continuous function that tends to $+\infty$ in every direction has a minimum.

5. Boosting with other losses, forests with few good features — 25 minutes — [20 pts]

Labels are $y \in \{-1, +1\}$ and the ensemble score is $F(x)$. The *exponential loss* is $\ell(y, F) = e^{-yF}$.

(a) Compute $g = \partial\ell/\partial F$ and $h = \partial^2\ell/\partial F^2$. [2 pts]

(b) Show that the population minimiser is $F^*(x) = \frac{1}{2}\log\frac{\eta(x)}{1-\eta(x)}$ with $\eta = \mathbb{P}(Y = +1 \mid x)$, and deduce how to turn a boosted score into a probability. [4 pts]

(c) Write the Newton leaf value with $\lambda = 0$ in terms of the weights $a_i = e^{-y_i F_i}$ and interpret the a_i. Compute it for a leaf of three examples with $y = (+1, +1, -1)$ and current scores $F = (0, 0, 3)$. Compare the influence of the third example with what it would have under the log-loss, and conclude which loss you would use when 10% of the labels are wrong. [5 pts]

(d) Regression boosting with the absolute loss $|y - F|$. Give the pseudo-residuals and explain why the Newton step cannot be used. Show that the leaf value minimising $\sum_{i \in q} |r_i - w|$ is a median, and compute it, and the mean, for residuals $(-3, -1, 0.5, 10)$. What does this buy? [4 pts]

A random forest has $d = 30$ features of which only $k = 3$ are informative.

(e) Show that the probability that at least one informative feature is among the m_{try} candidates

of a split is

$$1 - \binom{d-k}{m_{\text{try}}} \Big/ \binom{d}{m_{\text{try}}} = 1 - \prod_{i=0}^{k-1} \frac{d - m_{\text{try}} - i}{d - i}.$$

Compute it for $m_{\text{try}} = 5$ and $m_{\text{try}} = 10$, and explain why the default $m_{\text{try}} \approx \sqrt{d}$ can be a bad choice here. [5 pts]

6. From one flow to one host — 15 minutes — [10 pts]

(a) A detector has recall R and false-positive rate FPR; attacks have prevalence π. Show that its precision is

$$P = \frac{\pi R}{\pi R + (1 - \pi)\,\text{FPR}}.$$

Compute it for $R = 0.9$, FPR $= 0.01$ at $\pi = 0.1$ and at $\pi = 0.001$. Which of R, FPR, P are properties of the detector alone under prior shift, and which are not? [4 pts]

(b) A host is compromised (H_1) with prior 0.001. Each of its flows raises an alert independently, with probability 0.3 if compromised and 0.02 if clean. Show that after n flows with m alerts

$$\log \frac{\mathbb{P}(H_1 \mid \text{data})}{\mathbb{P}(H_0 \mid \text{data})} = \log \frac{0.001}{0.999} + m \log \frac{0.3}{0.02} + (n - m) \log \frac{0.7}{0.98}.$$

For $n = 20$, find the smallest m for which the host is more likely compromised than clean. [4 pts]

(c) Which model of Lecture 4 is this, which assumption did you use, and what happens to the answer of (b) if alerts come in bursts of three for the same underlying event? [2 pts]